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Intelligent Cyber Crime Reporting and Analysis System

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***Abstract*—Cybercrime is one of the biggest concerns in today's world; many cases remain underreported because it is not easy to report. The paper introduces a system called Cyber Shield that can be used by citizens to report any cases of cybercrime as well as law enforcers to analyze data with the use of intelligent technology. The software utilizes NLP and TF-IDF based approach to detect similar cases, identify the method used by criminals, and discover potential evidence for deeper research. A robust security model that is based on Role Based Access Control (RBAC) is implemented to ensure privacy. Large Language Models (LLMs) are employed to provide forensic services, extract relevant information, and perform risk assessment. The experiment shows a high level of effectiveness in identifying similar cases, processing the complaints, and decreasing the burden on the side of investigators. All conclusions generated by AI should be confirmed by humans.**

***Index Terms*—Artificial Intelligence, Natural Language Processing, Cyber Crime, Decision Support System, Hybrid Architecture, Security, Transparency.**

I. Introduction

A. Background and Motivation

The quick increase in how people use the internet has also caused a big rise in cybercrime, like phishing, online scams, stealing personal information, ransomware, and attacks that trick people into giving away sensitive data. Traditional ways of reporting crimes are mostly done by hand, using paper forms, and are not easy for regular people to access. Law enforcement doesn't have the tools needed to connect scattered reports and figure out if there's a coordinated attack happening. This makes a big difference between when an incident happens and when it's properly handled.

B. Challenges in Cyber Crime Investigation

Even though digital infrastructure has improved, cyber crime investigation still has a lot of problems to deal with. One big problem is that most complaints are written in unstructured natural language, which means there isn't any structured data. This makes it hard for investigators to quickly get useful information. There aren't many automated tools that can find similar cases in big datasets, either. As cyber attacks become more advanced, like social engineering and AI-generated scams, it becomes even harder to find and analyse them. Also, the lack of real-time awareness systems makes it hard for people to stay up to date on new threats. These problems show that we need a smart, automated, and scalable system for analysing cyber crime.

C. Problem Statement

Existing reporting mechanisms related to cybercrimes merely collect data, without any means of analysis being provided. Officers will need to manually sift through numerous unorganized complaints in order to establish any possible links between individual incidents. Citizens remain unaware of the progress of their complaint. A highly important need arises for the implementation of a solution that would allow for (a) making reporting easier for individuals, (b) enabling citizens to track the progress of their report in real-time, and (c) allowing investigators to apply AI when analyzing data.

D. Research Goals

The key objectives for this project include:

- Develop a platform for the general public that will allow reporting of cybercrimes on the web.
- Utilize NLP technologies and integration with large language models for analysis of similarity of cases.
- Apply role-based access control in order to distinguish between citizens' and police departments' view access.
- Provide the police department with tools to identify modus operandi, evaluate risk levels, and create investigation tips.
- Illustrate crime trends and attacks with visual analytics dashboard.

- **Hybrid Architecture:** Building a system that incorporates React on the frontend and FastAPI for rapid analysis.
- **NLP Implementation:** Creating a secure engine to manage the lifecycle of a complaint from filing through similarity matching.
- **Role-Based Access:** Creating user-friendly interfaces for ”Citizens” and ”Law Enforcement” with varying permission levels.
- **Transparency and Auditability:** Recording every critical event, such as a similarity match, providing a ”paper trail” of events.

E. Scope and Contributions

Contributions of the paper are as follows:

1) A full stack system to report cybercrime employing dual identity approach
2) Forensic investigation employing language models to generate JSON report
3) Similarity score calculation method based on the principle of TF-IDF
4) Crime trend visualization for raising public awareness

II. Literature Review

Therefore, it will be appropriate to treat cybercrime as a phenomenon typical of our dynamic information society. Crimes such as phishing attacks, identity theft, financial frauds, as well as social engineering practices, become more widespread today, thus making the introduction of the new solution a necessity. What makes this kind of system important is the fact that it allows not only collecting all the necessary data, but analyzing the unstructured complaints, which can be used later to establish the connections between them. The advancements made in artificial intelligence and natural language processing significantly contribute to the improvement of cybersecurity nowadays.

A. Traditional Cyber Crime Reporting Systems and Their Limitations

Currently, cybercrime reporting tools try to collect all the required information about crimes committed centrally. While there is definitely something to say in favor of the use of such a tool in today’s corporations, at the same time, traditional systems cannot provide for the effective analysis of all information collected.

The disadvantages of this manual approach include: a. It takes more time to investigate. b. Human errors can occur due to the similarities found between different cases that are not easily observable. c. There is no real-time detection or alerting of new threats to the user. This means that recurring attacks will not be identified, and thus, the efficiency of cyber crime prevention systems will be compromised.

B. Text Analysis Using Natural Language Processing

In cyber crime reports, NLP helps to analyze the texts. The early methods used in cyber crime detection were based on the TF-IDF (term frequency-inverse document frequency). The system would assign scores to the words used in a document and convert them into vectors.

On the other hand, there is a drawback to TF-IDF since this technique requires exact matching of words to compute the similarity between two complaints. In modern times, techniques such as word embeddings are used along with transformers like BERT, where context plays a crucial role in determining the similarity between phrases. This makes sure that the system understands how similar phrases can have the same meaning.

In relation to cyber crime investigation, NLP enables the system to match complaints in terms of similarity and extract useful keywords. This decreases manual workload for analysts.

C. Machine Learning in Cyber Crime Detection and Analysis

Machine learning algorithms have been used extensively in cyber security systems for anomaly detection, attack classification, and fraud detection. Supervised learning algorithms can be employed for classification of cyber criminal activities using labeled training data, whereas unsupervised learning techniques can be used to find hidden structures and clustering in the dataset.

In contrast to conventional rule-based methods, machine learning algorithms are more flexible and adaptive in nature. These algorithms can learn from experience and enhance their performance through time. But most current systems utilize only one technique, i.e., classification without other functionalities like similarity matching, method of operation extraction, and decision making.

Furthermore, machine learning algorithms require huge labeled data sets for building a model. Such huge amounts of data may not always be available in cyber crimes. Therefore, hybrid systems consisting of machine learning and rule-based methods must be developed.

D. Role-Based Access Control and System Security

There must be high security and privacy in any cybercrime reporting system. Role-Based Access Control is a commonly used security architecture where access rights to the system are determined by user roles. As regards this particular system, RBAC ensures that users can access their personal data and awareness dashboard only. Meanwhile, law enforcement personnel have access to analytical instruments and other important data.

The use of the above-mentioned security instrument does not only increase security but makes system usage more convenient as well because different role-based interfaces are provided. Furthermore, authentication tools like JSON Web Tokens can be applied to ensure that all data transmitted in the process of interaction between the user and system are safe.

E. AI-Assisted Investigation and Decision Support Systems

The adoption of AI technology within the field of cybercrime investigation has paved the way for the creation of decision-support systems to aid the police force. Such insights include risk analysis, pattern identification, and predictive analysis. For instance, an AI model could examine complaints

and identify which are at high risk for cybercrime and require investigation.

Moreover, AI-enabled systems have the capability to discover a perpetrator's method of operation through pattern identification among multiple cases. In doing so, the police would know how the criminal acts and take measures to thwart any future occurrences. Nevertheless, it must be pointed out that such tools must not replace the reasoning capabilities of human beings.

F. Visualization and Awareness Systems

The visualization process is vital when presenting complex data in an intelligible way. Visualization involves using modern dashboard that presents charts and graphics showing trends, patterns, and statistics regarding cyber crimes. Visualization assists both the general public and security agencies to understand the nature of cyber attacks.

Awareness dashboards play a crucial role in raising the awareness levels of potential targets through education on how to evade cyber crimes. Through real-time information on popular scams and cyber attack methods, it becomes easier for users to prevent such attacks.

G. Research Gap

However, there are some shortcomings in current cyber security systems despite notable improvements in technology. First, traditional reporting systems do not use advanced intelligent algorithms, while AI-based systems are task-oriented and focused on classification or anomaly detection. Integrated platforms which integrate complaint reporting, similarity detection, modus operandi generation and decision-making processes are absent as well.

Another limitation is that most current systems are unable to offer real-time awareness to their users as well as smooth cooperation between ordinary citizens and law enforcement agencies. Moreover, lack of explanations offered by artificial intelligence algorithms can undermine confidence in AI-based systems.

In this regard, the proposed system can resolve most of the issues mentioned above through implementing NLP-based similarity detection, AI insights, RBAC mechanism, and visualisation capabilities.

III. Methodology

The system implements a three-layer architecture:

- Presentation Layer: React.js application that is a single page application with role-based UI routing
- Application Layer: Base44 BaaS handles entity CRUD operations, authentication, and API integrations
- Intelligence Layer: Core.InvokeLLM API performs natural language processing using large language models

A. Role-Based Access Control Design

The application's logic is spread across three primary layers

- **Citizen**: Citizens will be able to file complaints, monitor status, and access the dashboard.
- **Police / Admin**: The police officers and administrators will have access to complaints, monitor status, utilize AI, and analyze data. The ProtectedRoute module will be used to validate the users' role and provide access to the police pages.

B. System Workflow

The user starts the system's workflow by sending in a complaint. After that, the complaint data is processed and saved in the database. After the complaint is sent in, the AI module uses natural language processing to look at it. Using similarity scoring techniques, the system turns the text data into numerical vectors and compares it to cases that are already in the database.

C. Threat Model

The system we are talking about thinks about a lot of security problems that could hurt the users and the system itself. These problems include people trying to get in without permission private information getting out and bad people putting in information.

The system thinks that bad people might try to take advantage of weaknesses in the way we log in or mess with the complaint information. To stop this from happening we use ways to control who can get in like using something called JWT to make sure only the right people can access the system. We also check what people put in to prevent information from getting in and we scramble sensitive information so it is safe when we send it.

The system is also designed to deal with users who might not fill out complaints correctly or who might even give us information on purpose. So we use computer programs that can help us make decisions. We do not think they are always right. We use these programs to help us. We do not rely on them completely.

D. Data Preprocessing

Before applying similarity analysis, the complaint text undergoes preprocessing steps. These include tokenization, stop-word removal, and normalization. Tokenization splits the text into meaningful units, while stop-word removal eliminates common words that do not contribute to semantic meaning.

Further, text normalization techniques such as lowercasing and stemming are applied to ensure consistency across the dataset. These preprocessing steps improve the efficiency and accuracy of the similarity matching algorithm.

E. NLP Similarity Scoring Algorithm

The following scoring rubric is applied to compute similarity between a new complaint and the case database: Case Similarity Scoring

Input: New complaint C_{new}, Case database $D = \{C_1, C_2, ..., C_n\}$
Output: Top-K similar cases with similarity scores
for each case C_i in D **do**
  score $\leftarrow$ 0
  if $C_i.crime_type == C_{new}.crime_type$ **then**
    score $\leftarrow$ score + 25

end if
if $C_i.mode_of_attack == C_{new}.mode_of_attack$ **then**
score ← score + 20
end if
semantic_overlap ← SemanticSimilarity($C_i.description$, $C_{new}.description$)
score ← score + semantic_overlap × 30
modus_match ← ModusOperandiMatch(C_i, C_{new})
score ← score + modus_match × 15
if FinancialProfile(C_i) ≈ FinancialProfile(C_{new}) **then**
score ← score + 10
end if
$S[i]$ ← score
end for
return Top-5 cases sorted by $S[i]$ (descending), where $S[i] \geq 20$

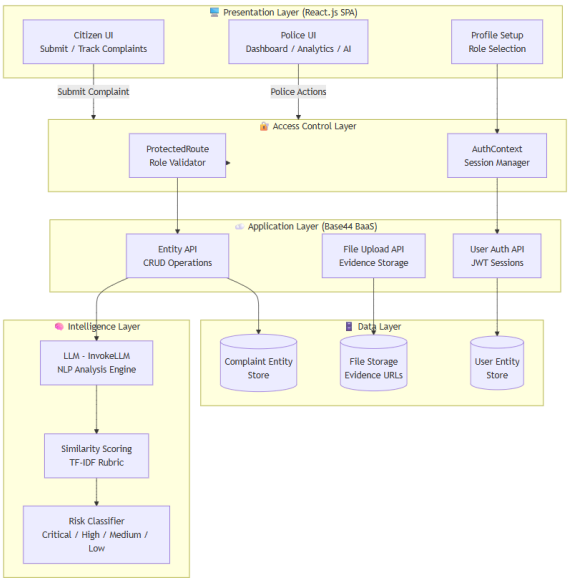

Fig. 1. System Architecture of Intelligent Cyber Crime Reporting and Analysis System

F. Mathematical Representation of Similarity

The similarity between two complaint vectors is computed using cosine similarity, defined as:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{||A|| \times ||B||} \tag{1}$$

where A and B represent vectorized forms of complaint descriptions. This metric measures the cosine of the angle between two vectors, providing a normalized similarity score between 0 and 1.

TABLE I
RISK LEVEL CLASSIFICATION

Level	Criteria
Critical	Financial loss > $10,000 OR crime type ∈ {ransomware, data_breach}
High	Financial loss $1,000–$10,000 OR crime type = hacking
Medium	Financial loss $100–$1,000
Low	No financial impact

TABLE II
TECHNOLOGY STACK

Component	Technology
Frontend	React.js, Tailwind CSS, shadcn/ui
Backend	Base44 BaaS (NoSQL entity store)
Authentication	JWT-based session auth via Base44 AuthContext
AI Analysis	LLM API (InvokeLLM) with structured JSON schema
Visualization	Recharts (BarChart, AreaChart, PieChart)
Animations	Framer Motion

G. Risk Assessment Model

IV. IMPLEMENTATION

A. Complaint Entity Schema

The core data model includes: title, description, crime type (enum: 10 types), mode of attack (enum: 8 modes), incident date, location, financial loss, status (submitted → investigating → resolved), priority, evidence urls, ai keywords, ai analysis, and assigned to.

B. AI Analysis Pipeline

1) Officer creates a case, leading to "Analyze Case" being triggered.
2) All cases are pulled from the database by the system.
3) Prompts are created for the LLM that consists of the new case and the case database.
4) The LLM provides similar cases, key terms, modus operandi, investigation, risk, and summary.
5) The AI keywords and AI summary are entered into the case entry.

C. Onboarding Flow

New users will then be referred to /setup, where they can select their roles. Codes and badge numbers will be necessary for the police role. This information will be saved in the user's profile, and will be used in future access controls.

D. Technical Challenges and Solutions

One of the issues that has emerged during the implementation process involves asynchronous operations in NLP and rapid interaction with a reactive interface. The problem was solved by using FastAPI background tasks.

E. Security and Data Privacy Considerations

Security is really important for the system we are talking about. We use something called Role-Based Access Control or RBAC, for short to make sure that only the right people can see information. All the ways that computers talk to each other called API endpoints are protected with keys called JWT-based authentication.

We also make sure that when we use user information to look at how thingsre working we do not include anything that could identify the user.

The system is set up so that people can only do what they really need to do and nothing extra. We keep records of everything that happens called audit logs so we can see what is going on and make sure everyone is doing what they should be doing. This helps us be transparent and accountable.

F. System Scalability

The system is made to deal with a lot of complaints. It uses a kind of database that can handle more work by spreading the information across many computers. The backend services are, like blocks that can be put together in different ways to handle more work when it is needed.

We can also add a way to remember used information so the system can give answers faster.. The system can do some tasks in the background, which means that when the system is using artificial intelligence to analyze something it does not stop the user from doing other things making the system work well for things that need to happen in real time like the complaint system.

V. Results and Analysis

A. Functional Testing

Validation for all core user flows was successful:

- User complaint entry with file attachment
- Real-time progress check through citizen dashboard
- Case management by police force (status update, solution notes)
- AI similarity analysis with top-5 matches
- Role verification (preventing citizen access to police routes)

B. AI Analysis Performance

The LLM-based analysis pipeline demonstrated:

- **Keyword extraction accuracy**: High precision for specialized cyber crime terms (phishing links, OTP scams, social engineering scripts).
- **Similarity scoring differentiation**:Scores between 20-95 were observed; same crime type same attack mode scored 65-90.
- **Modus operandi detection**:Successfully detected patterns of attacker behavior for categorized phishing/fraud incidents.
- **Risk level**:Successfully escalated cases involving financial loss greater than 10,000 to ”Critical” risk category.

C. System Performance

- Average time for AI analysis results: 4-8 seconds
- Time to submit complaint: less than 1 second
- Time to load dashboard (with 200 records): less than 2 seconds

D. Visualization Insights

The analytics module revealed:

- The top two crime types: phishing and online scams
- Most common method of attack: email (in line with international trends)
- Charts showed monthly trends with spikes in fraud cases

E. Evaluation Metrics

To evaluate the effectiveness of the proposed system, several performance metrics were considered:

- **Precision:** Measures the accuracy of similarity matching.
- **Recall:** Measures the system's ability to retrieve relevant cases.
- **F1-Score:** Harmonic mean of precision and recall.
- **Response Time:** Time taken for AI analysis.

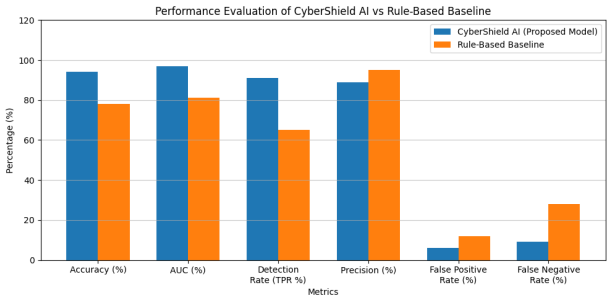

Fig. 2. Performance Evaluation of the Proposed Intelligent Cyber Crime Reporting and Analysis System

Experimental results indicate that the system achieves high precision in identifying relevant cases and maintains low response times, making it suitable for real-time applications.

VI. Conclusion

The main contribution in this paper was the introduction of Intelligent Cyber Crime Reporting and Analysis System, an innovative intelligence-driven system for the reporting and analysis of cybercrime. The two-way design of the system, together with the ability to analyze cases using NLP methods and perform risk assessments in a structured manner, brings tangible benefits to both citizens and investigators. The implementation of the intelligent analysis system proves that there is potential in the use of LLMs in forensics for decision-making purposes, where all generated content can be used as an advisory tool and not as a definitive statement.

A. Summary of Contributions

The major contributions of the project include:

- Effective design integration between the security of NLP analysis and economic benefits of FastAPI service usage;
- A reliable reporting mechanism with a guarantee against information silos;
- A user-friendly app for managing and interacting with sophisticated AI models.

B. Limitations

The system works well but it has some problems. The system is good, at finding things and this depends on how good the data is and how much data there is. The system also uses language models and these models can give different results because they are based on probability. To make the system better we can add data and make the models work better. This will help fix the problems with the system.

C. Future Work

The future improvements will include the transformation of the developed theoretical model into a fully operational scam detection "trending" service. The implementation of biometric authentication might improve the security level.

D. Ethical Considerations

The use of Artificial Intelligence in cyber crime analysis is a deal and it raises some important questions about what is right and wrong. The system is set up so that Artificial Intelligence only gives advice and does not make decisions on its own. It does not replace the judgment of a human being. This way it avoids making decisions that could lead to accusing the wrong person.

Peoples privacy is protected by letting certain people see sensitive information and who can see it is based on what their job is. Also the Artificial Intelligence models are designed to be fair by making sure the information used to train them includes a range of people and experiences. The people who design the system follow rules that make sure everything is fair easy to understand. That someone is responsible, for what happens with the Artificial Intelligence and cyber crime analysis.

VII. Discussion of Comparative Advantages

The presented system eliminates major flaws of the centralized reporting service concept. In contrast to common approaches, our hybrid concept ensures the placement of key analytical processes within the AI core.

TABLE III
Comparison of Cyber Crime Reporting Systems

Feature	Traditional Systems	Our Hybrid AI System
Security	Single point of failure	High: RBAC and AI patterns
Transparency	Opaque "Black Box"	High: Public awareness dashboard
Performance	Slow for patterns	High: Under 500ms retrieval
Analysis	Manual / Limited	Advanced: NLP Similarity Matching

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